Academic Year: 2026-27 **Semester: V**
Class/Branch: TE/DS **Subject: Agile**

Assignment No.2

Course Outcome(CO)/Lab Outcome(LO):

LO1	To apply Agile and Scrum methodologies using Jira for sprint planning, backlog management, and progress tracking, and integrate Agile workflows with DevOps pipelines.
LO2	To apply Git for distributed version control, source code management, branching and merging strategies, and collaborative team development.
LO3	To build and manage automated CI/CD pipelines using Jenkins and GitHub Actions for building, testing, packaging, and deploying applications.
LO4	To Containerise, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and provision cloud infrastructure using Terraform.
LO5	To implement ML lifecycle management and observability using MLflow, Airflow, AWS Lambda, Prometheus, and Grafana.
LO6	To Design, integrate, and demonstrate an end-to-end DevOps pipeline covering planning, build, test, deployment, and monitoring.

Q 1 : Implement a DevOps CI/CD pipeline for an application using GitHub Actions or Jenkins. Configure automated build and testing, containerize the application using Docker, and deploy it using Docker Compose. Demonstrate the execution of the pipeline with suitable screenshots. [LO4, L3]

Q 2 : Implement the Machine Learning lifecycle using MLflow by tracking experiments, logging parameters and metrics, managing model versions, and deploying the trained model. Demonstrate basic application monitoring using Prometheus and Grafana. [LO5, L3]

Q 3: Design, integrate, and demonstrate an end-to-end DevOps pipeline for a selected application by integrating Agile planning, Git-based version control, automated CI/CD, application deployment, and monitoring. Present the complete workflow using an architecture diagram and execution evidence. [LO6, L6]